

Varun K R

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WORK EXPERIENCE

AI Engineer

10/2025 – 04/2026

Iolite Technologies Pvt LTD

Bengaluru, Karnataka

- Developed a production NL-to-SQL pipeline for examination reporting using FastAPI, LangChain, MySQL, and ChromaDB, enabling natural language queries over 500K+ student records and reducing report generation time from 2-3 hours to 8-10 seconds for 1500+ faculty members.
- Implemented a RAG-based entity resolution pipeline to accurately identify courses, programs, departments, and campuses from the university database, improving NL-to-SQL query precision to 99%.
- Developed a ReAct-based booking assistant that handled booking, cancellation, and listing requests through ERP API orchestration, slot extraction, and Redis-backed conversational state management across 6 campuses, reducing venue booking time by 70% through natural language interaction.
- Collaborated on deployment and monitoring-ready infrastructure using AWS, Docker, Langfuse, and Python async services.

EDUCATION

Bachelor of Computer Applications

07/2023-07/2026

Christ (Deemed to be University)

Delhi, NCR

SKILLS

Programming Languages: Python, C++, JavaScript, SQL

Artificial Intelligence & Machine Learning: LLMs, AI Agents, RAG Pipelines, Prompt Engineering, ReAct Agents, Tool Calling, NLP, Vector Databases

Frameworks & Libraries: FastAPI, LangChain, LangGraph, Pydantic, HTTPX, SQLAlchemy, DSPy

Databases & Storage: MySQL, ChromaDB, PostgreSQL, Redis

DevOps & Tools: Docker, Git, Linux, AWS, Langfuse

PROJECTS

Booking Agent

02/2026 – 04/2026

ReAct Based Agentic Intelligent Campus Booking System

- Built an AI-powered conversational booking system for a university ERP, allowing users to book, cancel, and query venues through natural language while reducing clicks, navigation, and manual data entry.
- Developed a multi-agent LangChain workflow with ReAct agents that verified entities, checked availability, and autonomously executed ERP API operations across 6 campuses with stateful conversation management using Redis

Examination Reports

10/2025 – 02/2026

AI-Powered Natural Language to SQL Analytics Platform

- Built an NL-to-SQL conversational analytics platform using FastAPI, React, and LangChain, translating natural language into verified MySQL queries for 500K+ student records with chart generation, auto-correction, and observability using Langfuse.
- Implemented Redis-backed conversation memory and ChromaDB-based RAG entity resolution to resolve courses, programs, departments, and campuses accurately across multi-turn queries.
- Implemented schema-aware query generation with two-stage SQL verification, leveraging database metadata to select relevant tables and columns, reducing hallucinations by 60% and enabling faculty to generate reports without SQL expertise